# Improving Fundamental Stock Selection with a Model Ensemble

Columbia University  Quanjing He (qh2305)

## 1. Introduction

This study improves the fundamental stock selection baseline in FinRL-Trading. The original workflow trains Random Forest, XGBoost, and LightGBM models by sector, selects the model with the lowest validation mean squared error (MSE), and buys the top 25% of stocks by predicted return. After reproducing the baseline, I found weak ranking performance and several data-timing and backtest limitations. I first corrected the evaluation framework, then tested changes to selection, weighting, rebalancing, and model combination. The final strategy, S9, averages predictions from all three models and uses weekly rebalancing. I compare its net performance with a corrected baseline and the SPY and QQQ ETFs.

## 2. Data and Bias Controls

I use the public FinRL-Trading fundamental, price, and historical S&P 500 constituent datasets. The prepared fundamental sample covers 42 quarters from June 2015 to September 2025, with 19,007 observations across 674 tickers and 11 GICS sectors. The historical membership filter reduced 21,299 eligible fundamental rows to 19,007, removing 2,292 rows (10.76%) and 18 tickers entirely. The sample retains past constituents such as ALXN, ABMD, and ATVI rather than using only current S&P 500 members. Trading is evaluated from October 2, 2020, to September 30, 2025.

To reduce look-ahead bias, I treat a financial observation as available no earlier than the later of its assigned quarter-end and recorded reporting date. I also require training and validation return labels to be realized before each model decision. Signals enter the portfolio no earlier than the next trading day after they become available. I check historical index membership and valid prices on each trading day, and exclude quarterly adjusted closing prices from the 52 model features. The timing audit found no label-date violations in 220 sector-quarter model windows, and no historical membership violations among 90,096 recorded S9 stock-day holdings. These are best-effort point-in-time controls; source filing timestamps and underlying ratio construction cannot be fully verified.

## 3. Strategy Design and Experiments

The original baseline selected XGBoost in 90 of 231 sector-quarter windows, Random Forest in 80, and LightGBM in 61. Its mean quarterly Rank IC was only 0.0032. Because the original baseline uses intraday stock returns while the ETF benchmarks use close-to-close returns, I built a corrected framework with adjusted open-to-next-open returns for every strategy and benchmark. Stock portfolios pay 0.1% of absolute turnover, including changes from weight drift. The corrected baseline rebalances the top 25% portfolio daily.

I tested nine modifications. More concentrated selection (S1) and a 60-day trend filter (S2) did not improve validation returns. Inverse-volatility weighting (S3) and the ML–momentum blend (S6) did not outperform the corrected baseline. Monthly rebalancing (S5) improved validation performance but performed worse during development. Weekly rebalancing (S4) improved validation CAGR from 15.45% to 17.46%. Sector-relative ranking (S7) and Rank-IC-based model selection (S8) offered mixed results. S9 achieved the highest development and validation CAGR among the tested candidates.

*Table 1. Strategy comparison during development and validation. CAGR is annualized.*

| Strategy | Development CAGR | Validation CAGR |
|---|---|---|
| Corrected Baseline | 13.14% | 15.45% |
| S1 Top 10% | 13.26% | 12.98% |
| S2 Trend Filter | 2.31% | 3.03% |
| S3 Inverse Volatility | 10.67% | 14.05% |
| S4 Weekly Rebalance | 13.02% | 17.46% |
| S5 Monthly Rebalance | 8.49% | 16.87% |
| S6 Hybrid Momentum | 7.37% | 15.03% |
| S7 Sector-Relative | 14.34% | 13.97% |
| S8 Rank-IC Selection | 14.45% | 16.30% |
| **S9 Model Ensemble** | **15.51%** | **18.38%** |
| SPY | 13.53% | 26.64% |
| QQQ | 13.83% | 28.23% |

For S9, I retrain the three models on the same point-in-time rolling samples and average their predicted returns. I rank eligible stocks across sectors, select the top 25%, set equal target weights each week, and allow weights to drift between rebalances. Positions that become ineligible are removed; unused capital stays in cash. The ensemble replaces single-model selection but keeps the trading rules fixed for the final evaluation.

I also examined permutation feature importance for Sector 45 in the September 2024 model window. Book value per share (BPS), fixed asset turnover, and operating cash flow to sales ranked highest. However, the mean validation Rank IC was negative (−0.0523), so this single-window result does not establish stable predictive importance.

## 4. Backtest Results

Table 2 reports results over October 2020–September 2025. S9 reached a net final NAV of 2.4642 (a 146.42% cumulative return), compared with 2.1438 for SPY and 2.2353 for QQQ. Its CAGR was 19.79%, exceeding SPY by 3.29 percentage points, and its Sharpe ratio was 1.0525. Its maximum drawdown of −30.56% was nevertheless worse than SPY's −26.29%.

*Table 2. Full-period performance. Initial NAV = 1; Sharpe assumes a zero risk-free rate.*

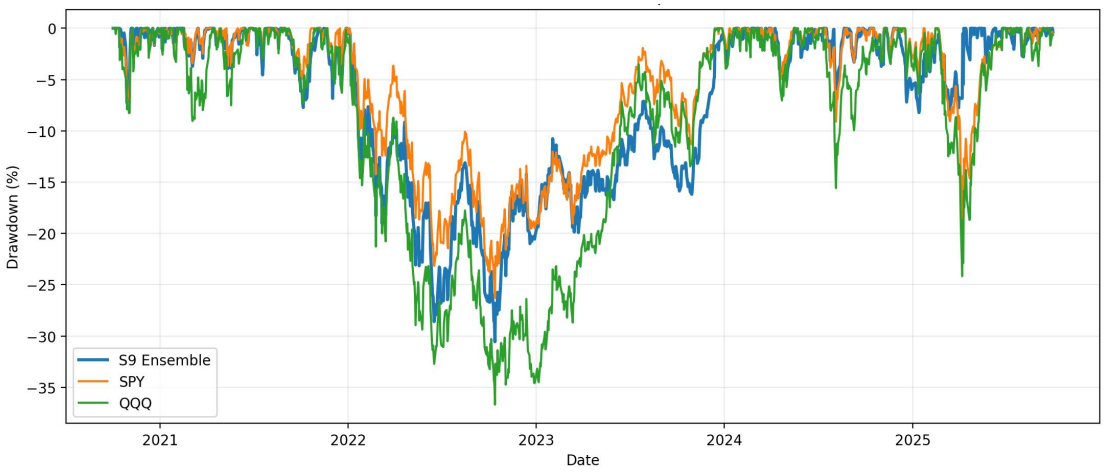
| Portfolio | Final NAV | CAGR | Volatility | Sharpe | Max DD |
|---|---|---|---|---|---|
| **S9 Ensemble** | **2.4642** | **19.79%** | **18.94%** | **1.053** | **-30.56%** |
| Corrected Baseline | 1.9482 | 14.29% | 19.61% | 0.782 | -32.54% |
| SPY | 2.1438 | 16.50% | 17.48% | 0.965 | -26.29% |
| QQQ | 2.2353 | 17.48% | 22.97% | 0.819 | -36.69% |

S9 did not outperform the ETFs in every period. During 2024 validation, its CAGR was 18.38%, versus 26.64% for SPY and 28.23% for QQQ. In the January–September 2025 test interval, S9 earned 30.87% cumulatively, versus 13.46% for SPY and 16.80% for QQQ. S9 averaged 0.0877 daily turnover, compared with 0.1094 for the corrected baseline. The ETF benchmarks are modeled as buy-and-hold positions without the same rebalancing transaction costs.

*Figure 1. Cumulative NAV: S9, corrected baseline, SPY, and QQQ.*



*Figure 2. Portfolio drawdowns: S9, SPY, and QQQ.*




## 5. Robustness, Limitations, and Conclusion

Historical price data were unavailable for 93 tickers, affecting 868 candidate signals. I recovered prices for two additional ticker-format aliases (BF.B and BRK.B) but could not reconstruct the missing histories. I also flagged 30 financial records with unavailable or inconsistent filing-date metadata. Removing all 30 flagged signals from portfolio selection reduced full-period S9 NAV only from 2.4642 to 2.4556, still above SPY at 2.1438. This sensitivity test did not retrain the models, so it does not rule out the influence of these records on fitted predictions.

The remaining price gaps may introduce selection or survivorship bias, and the reporting dates may not match when every financial ratio first became public. The validation year was used repeatedly while developing strategies; the 2025 baseline results were also observed before the final S9 assessment. For these reasons, the 2025 test is not a fully untouched blind test. The results support S9 as the best candidate in this experiment, not as proof of reliable live outperformance. A stronger follow-up would reconstruct historical price coverage and actual feature release dates, then test the frozen strategy on a later untouched period.